Supplementary material

Long-term trends in enterohaemorrhagic *Escherichia coli* notifications in the Republic of Korea: a negative-control analysis of 24 years of nationwide surveillance

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Supplementary Methods

Trend model. Annual national notification rates were modelled with a negative-binomial generalised linear model (NB2 parameterisation; log link, log-population offset, linear calendar-year term), yielding a multiplicative rate ratio per year and the dispersion parameter α (variance = μ + αμ²). Confidence intervals for the rate ratio are Wald intervals on the log scale computed from the maximum-likelihood standard error of the fitted year coefficient; the interval for α is a Wald interval on the log-α scale. Overdispersion was tested with the likelihood-ratio statistic 2(ℓ_NB − ℓ_Poisson) against the half-χ^2_0 : χ^2_1 mixture reference (Supplementary Table S3). Denominators are year-end KOSIS resident-registration populations aggregated from the 229 sigungu series after removal of the duplicate city-plus-district records described in the Methods; the resulting national totals reproduce the published KOSIS annual figures exactly. The national total rose by 1.5% between 2009 and 2010, against 0.4% to 0.6% in adjacent years, mainly because about 466,000 previously deregistered residents were restored to the resident register as residence-unknown registrants in October 2010. Inflating the 2001–2009 denominators by this amount changed the EHEC trend estimate from 9.75% to 9.82% per year (95% CI 7.27 to 12.43), a difference below the reported precision. Three pre-specified sensitivity analyses re-fitted the same model to 2010–2024, to the series excluding 2020, and to domestic cases only; a post hoc fourth analysis re-fitted it to 2007–2024, the period after the transient early excess identified by the breakpoint search (Supplementary Table S8).

Structural change. Continuous piecewise-linear negative-binomial models of the form $\log(\text{rate}_t) = \beta_0 + \beta_1 t + \Sigma_b \beta_b \max(0, t - bp_b)$ were fitted exhaustively for 0–3 breakpoints, with the dispersion fixed at the constant-slope estimate (α = 0.155 for 2001–2024; α = 0.028 for 2007–2024). Breakpoints were placed so that each segment spanned at least h_min annual observations; h_min = 3 was the primary specification and h_min = 2 and 4 were sensitivity analyses (Supplementary Table S9). ΔBIC denotes BIC(breakpoint model) − BIC(constant-slope model), so negative values favour a breakpoint; on the Kass and Raftery scale ΔBIC ≤ −2 corresponds to positive, ΔBIC ≤ −6 to strong and ΔBIC ≤ −10 to very strong evidence, and a break was declared only at the pre-specified criterion ΔBIC < −6. In the first revision, only the single-breakpoint comparison was re-run after the dispersion was corrected; the two- and three-breakpoint models had been fitted with the dispersion of the original pipeline. The full search reported in Supplementary Table S9 corrects this. The power simulation generated 1,000 24-year series per effect size at the empirical dispersion (α = 0.16), with detection defined as ΔBIC < −6 between the best single-breakpoint model and the constant-slope model (Supplementary Table S6; Supplementary Figure S1).

Spatial analysis. The baseline weight matrix was Queen contiguity with a two-nearest-neighbour fallback for Jeju island, row-standardised; global Moran's I significance used 9,999 free Monte-Carlo permutations. Twelve alternative schemes (Queen with Jeju isolated, binary-weight variants, k-nearest-neighbour with k = 1, 3, 4 and 5, and 100–300-km distance bands) were examined with 999 permutations each; k = 2 is not listed separately because it coincides with the baseline Queen + KNN-2 specification (Supplementary Table S1). Local indicators of spatial association used 999 conditional permutations with Benjamini–Hochberg false-discovery-rate adjustment across the 16 provinces (Supplementary Table S7). Empirical Bayes smoothing used Marshall's method-of-moments shrinkage of sigungu-level rates toward the global mean before population-weighted re-aggregation to provinces (Supplementary Table S2).

Age standardisation was direct, to the 2005 Korean Standard Population eight-band weights, computed from the annual national notification extract (Supplementary Table S4).

Supplementary Table S1. Spatial autocorrelation under 12 weighting schemes (Moran's I, 999 permutations per scheme)

Scheme	I (domestic)	p (dom.)	I (total)	p (total)
Queen (Jeju isolated, row-std)	0.176	0.2270	0.172	0.2390
Queen (Jeju isolated, binary)	0.157	0.2710	0.155	0.2680
Queen + KNN-2 Jeju (row-std)	0.448	0.0010	0.440	0.0010
Queen + KNN-2 Jeju (binary)	0.324	0.0110	0.319	0.0120
KNN k=1 (row-std)	0.074	0.7100	0.060	0.7610
KNN k=3 (row-std)	0.108	0.5150	0.103	0.5480
KNN k=4 (row-std)	0.088	0.5820	0.084	0.5940
KNN k=5 (row-std)	0.070	0.6100	0.066	0.6260
Distance band 100km (row-std)	0.067	0.7480	0.062	0.7690
Distance band 150km (row-std)	0.022	0.8890	0.020	0.9080
Distance band 200km (row-std)	0.188	0.0070	0.186	0.0080
Distance band 300km (row-std)	−0.001	0.9970	−0.003	0.9950

Supplementary Table S2. Provincial cumulative cases and Empirical Bayes (Marshall, 1991) sido aggregate

Panel A: Provincial cumulative cases and mean annual rate.

Province	Total	Domestic	Imported	Mean pop	Mean annual notification rate per 100,000
Gwangju	395	392	3	1,436,040	1.146
Jeju	90	90	0	607,195	0.618
Jeonnam	147	146	1	1,914,206	0.320
Ulsan	79	78	1	1,120,787	0.294
Daegu	140	131	9	2,473,386	0.236
Incheon	141	137	4	2,807,346	0.209
Daejeon	70	66	4	1,475,511	0.198
Gyeongbuk	118	116	2	2,676,582	0.184
Chungnam	83	81	2	1,977,786	0.175
Gyeonggi	495	475	20	12,522,082	0.165
Gangwon	60	58	2	1,532,222	0.163
Seoul	352	328	24	9,975,073	0.147
Jeonbuk	63	60	3	1,858,644	0.141
Gyeongnam	100	95	5	3,063,195	0.136
Chungbuk	42	39	3	1,515,779	0.115
Busan	73	70	3	3,519,439	0.086

Sejong (17 cases over 2001–2024) is excluded from Panel A because complete annual denominators are unavailable; it is shown in Panel B, which is computed from the municipal series, but is not included in any Moran's I statistic. Panel A column totals therefore sum to 2,448 cases (2,362 domestic, 86 imported) rather than the national totals in main-text Table 1.

Panel B: EB-smoothed (Marshall, 1991) sido-level aggregate vs raw rates and Moran's I comparison.

Province	# sigungu	Domestic cases	Raw domestic rate per 100,000	EB-smoothed rate per 100,000
Busan	16	70	0.083	0.089
Chungbuk	11	39	0.105	0.109
Chungnam	15	81	0.168	0.168
Daegu	9	131	0.219	0.219
Daejeon	5	66	0.186	0.186
Gangwon	18	58	0.158	0.161
Gwangju	5	392	1.137	1.090
Gyeongbuk	22	116	0.182	0.181
Gyeonggi	31	475	0.165	0.166
Gyeongnam	18	95	0.121	0.125
Incheon	10	137	0.203	0.203

Jeju	2	90	0.618	0.603
Jeonbuk	14	60	0.135	0.137
Jeonnam	22	146	0.318	0.296
Sejong	1	16	0.350	0.340
Seoul	25	328	0.137	0.139
Ulsan	5	78	0.290	0.284

Moran's I (Queen + KNN-2 baseline, 9,999 permutations): raw I = 0.448 (p = 0.0003) for the domestic series used for smoothing (0.440 for all notifications); EB-smoothed I = 0.440 (p = 0.0004). Empirical Bayes smoothing was applied to the domestic sigungu series (df1), which is the only extract available at sigungu level; the raw comparison rate in Panel B is the corresponding domestic rate. The south-western gradient was not materially changed by this smoothing procedure.

Supplementary Table S3. Negative-binomial dispersion α (95% CI)

Series	n years	Total events	$\hat{\alpha}$	95% CI	Pearson χ^2/df (Poisson)	LR vs Poisson	p (one-sided)
total	24	2,465	0.155	0.080–0.297	13.49	150.6	< 0.001
domestic	24	2,378	0.162	0.085–0.309	14.47	163.9	< 0.001
imported	24	87	1.011	0.449–2.276	4.94	50.8	< 0.001

All negative-binomial regressions were fit on n = 24 annual case totals against the corrected resident-registration denominators. The Pearson χ^2/df column is the diagnostic of the Poisson fit and motivates the negative-binomial family; under the fitted negative-binomial models the corresponding ratio is close to unity. The likelihood-ratio statistic of 150.6 (total) and 163.9 (domestic) reflects the magnitude of the variance mismatch between Poisson and the empirical annual count distribution, not a large sample size.

Supplementary Table S4. Age-standardised national notification rate (KSP-2005)

Year	ASR (KSP-2005) per 100,000	ASR 95% CI
2001	0.019	0.007–0.032
2002	0.013	0.003–0.022
2003	0.088	0.062–0.114
2004	0.222	0.180–0.263
2005	0.084	0.058–0.110
2006	0.076	0.051–0.100
2007	0.079	0.053–0.104
2008	0.108	0.079–0.137
2009	0.127	0.094–0.160
2010	0.119	0.086–0.151
2011	0.155	0.118–0.192
2012	0.133	0.099–0.168
2013	0.139	0.104–0.175
2014	0.256	0.208–0.305
2015	0.160	0.122–0.198
2016	0.223	0.179–0.267
2017	0.303	0.251–0.355
2018	0.259	0.211–0.308
2019	0.312	0.258–0.366
2020	0.709	0.622–0.796
2021	0.431	0.362–0.499
2022	0.548	0.469–0.626
2023	0.536	0.458–0.614
2024	0.697	0.605–0.788

Direct age standardisation to the 2005 Korean Standard Population, computed from the annual national notification extract and the corrected resident-registration denominators. The age-standardised series shows the same upward trajectory as the crude rate in main-text Table 1; age standardisation does not remove the trend, confirming that the rise in EHEC notifications is not an artefact of changing age structure.

Supplementary Table S5. Negative-control comparison of three notifiable enteric infections, 2001–2024

Infection	Cases 2001	Cases 2024	Total	NB GLM (%/yr, 95% CI)
EHEC	10	274	2,465	+9.8 (+7.2, +12.4)
Shigellosis	817	41	5,682	−12.1 (−14.4, −9.6)
Typhoid fever	384	34	3,535	−7.3 (−9.6, −4.9)

National annual case totals (KDCA df2 extract) and notification rates per 100,000 population against the same resident-registration denominators used for EHEC. Estimates are from the negative-binomial GLM with log-population offset, with Wald 95% confidence intervals from the fitted model. Modern multiplex gastrointestinal PCR panels co-detect Shiga toxin-producing *E. coli* and *Shigella*; the concurrent decline of shigellosis and typhoid makes a system-wide, non-pathogen-specific expansion of culture-independent testing unlikely to be the sole explanation for the EHEC rise, but does not exclude pathogen-specific changes in diagnostic sensitivity for STEC. See main-text Figure 3.

Supplementary Table S6. Single-breakpoint detection power

Negative-binomial breakpoint power simulation at the dispersion used for the simulation (α = 0.16; the corrected empirical value, 0.155, is immaterial at this precision); n = 24 annual observations; 1,000 simulated series per effect size; detection = ΔBIC < −6 between the BIC-best 1-breakpoint and 0-breakpoint NB GLMs). Panel A: a continuous post-breakpoint additional slope change imposed from the series midpoint. Panel B: a single-year multiplicative step shock. The series is well-powered for sustained slope changes ≥25%/year but, as expected for 24 annual points, insensitive to isolated single-year shocks.

Panel A: continuous post-breakpoint slope change

Additional slope (%/yr)	Detection probability	n simulations
0	0.8%	1000
25	93.6%	1000
50	100.0%	1000
75	100.0%	1000
100	100.0%	1000
150	100.0%	1000

Panel B: single-year multiplicative step shock

Step multiplier (×)	Detection probability	n simulations
1	1.9%	1000
1.5	1.5%	1000
2	3.4%	1000
3	11.4%	1000
5	40.3%	1000

Supplementary Table S7. LISA local Moran's I with Benjamini-Hochberg FDR adjustment

Local Moran's I (LISA) for the 16 provinces on cumulative notification rates (Queen + KNN-2; 999 conditional permutations), with Benjamini-Hochberg FDR-adjusted q-values. Gwangju, Jeju and Jeonnam are high-high under raw permutation (p = 0.006, 0.008 and 0.040) but none survives FDR adjustment (q ≥ 0.064); the south-western signal is therefore supported primarily by the significant global Moran's I rather than by province-level local significance, and no province should be described as a statistically confirmed local cluster.

Province	Rate per 100,000	Local I	Quadrant	p (raw)	q (BH-FDR)	FDR-sig
Gwangju	1.146	2.646	HH	0.006	0.064	no
Jeju	0.618	2.448	HH	0.008	0.064	no
Jeonnam	0.320	0.180	HH	0.040	0.213	no
Chungbuk	0.115	0.237	LL	0.246	0.587	no
Ulsan	0.294	−0.047	HL	0.261	0.587	no
Gyeonggi	0.165	0.176	LL	0.294	0.587	no
Gangwon	0.163	0.191	LL	0.339	0.587	no
Daejeon	0.198	0.140	LL	0.348	0.587	no
Chungnam	0.175	0.170	LL	0.356	0.587	no
Gyeongbuk	0.184	0.120	LL	0.367	0.587	no
Incheon	0.209	0.108	LL	0.419	0.609	no

Jeonbuk	0.141	0.168	LL	0.498	0.633	no
Daegu	0.236	0.059	LL	0.514	0.633	no
Seoul	0.147	0.158	LL	0.657	0.711	no
Gyeongnam	0.136	0.125	LL	0.667	0.711	no
Busan	0.086	0.157	LL	0.782	0.782	no

Supplementary Table S8. Sensitivity analyses of the long-term trend

The same negative-binomial model (log link, log-population offset, linear year term) re-fitted to restricted series. Restricting the analysis to the period after the main transition to culture-independent diagnostics does not attenuate the rise; excluding the single largest outbreak year and restricting to domestically acquired cases leave the estimate essentially unchanged. The 2007–2024 row is a post hoc analysis that follows the breakpoint search (Supplementary Table S9). Confidence intervals are Wald intervals on the log scale from the fitted model.

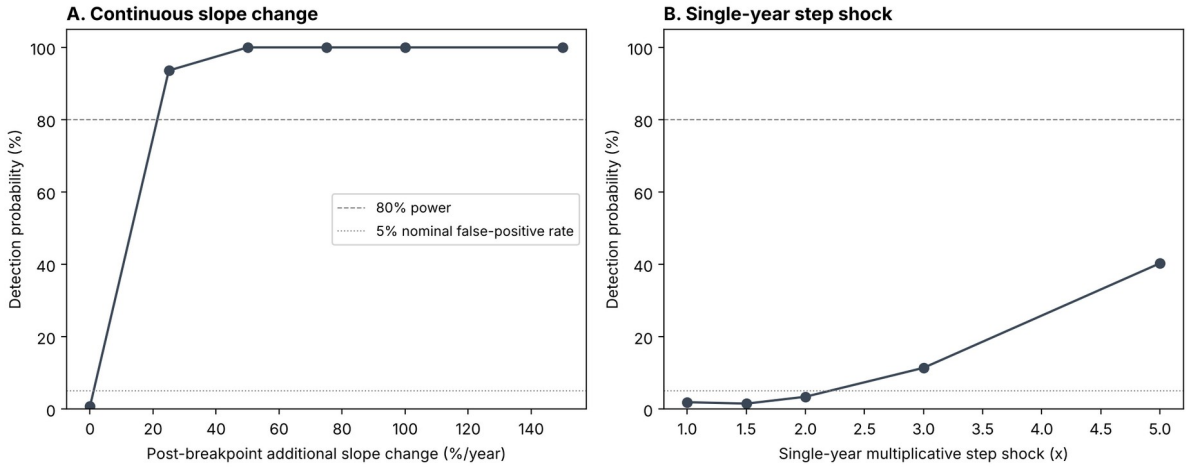
Analysis	**n years**	**Cases**	**Change per year (%, 95% CI)**
Full series, 2001–2024 (primary)	24	2,465	+9.8 (+7.2, +12.4)
Modern diagnostic era, 2010–2024	15	2,068	+12.4 (+9.8, +15.1)
Excluding the 2020 outbreak year	23	2,195	+9.7 (+7.0, +12.4)
Domestic cases only, 2001–2024	24	2,378	+9.6 (+7.0, +12.2)
2005–2024 (post-2004 candidate break)	20	2,293	+11.2 (+9.6, +12.9)
2007–2024 (post hoc; after the transient early excess)	18	2,216	+11.6 (+9.7, +13.6)

Supplementary Table S9. Exhaustive breakpoint search by minimum segment length

ΔBIC = BIC(best model with the stated number of breakpoints) − BIC(constant-slope model); negative values favour breakpoints and the pre-specified criterion is ΔBIC < −6. Years in parentheses are the breakpoint locations of the best model. Dispersion fixed at the constant-slope estimate of each series. h_min = 3 is the primary specification. Over 2001–2024 the two-breakpoint model (2004, 2007) meets the criterion; it captures a transient excess around 2004, when 81 of 110 notifications came from one district (Nam-gu) of Gwangju. Over 2007–2024 no breakpoint model is favoured (all ΔBIC > 0).

Series	**h_min**	**1 breakpoint: ΔBIC (year)**	**2 breakpoints: ΔBIC (years)**	**3 breakpoints: ΔBIC (years)**
2001–2024	2	−5.36 (2003)	−12.31 (2004, 2006)	−9.23 (2004, 2006, 2008)
2001–2024	3	−4.34 (2004)	−11.02 (2004, 2007)	−7.98 (2004, 2007, 2020)
2001–2024	4	+1.52 (2005)	−1.71 (2005, 2009)	+1.25 (2005, 2009, 2020)
2007–2024	2	+2.08 (2012)	+2.34 (2018, 2020)	+3.30 (2018, 2020, 2022)
2007–2024	3	+2.08 (2012)	+2.76 (2015, 2020)	+5.60 (2010, 2015, 2020)
2007–2024	4	+2.08 (2012)	+2.76 (2015, 2020)	+5.60 (2011, 2015, 2020)

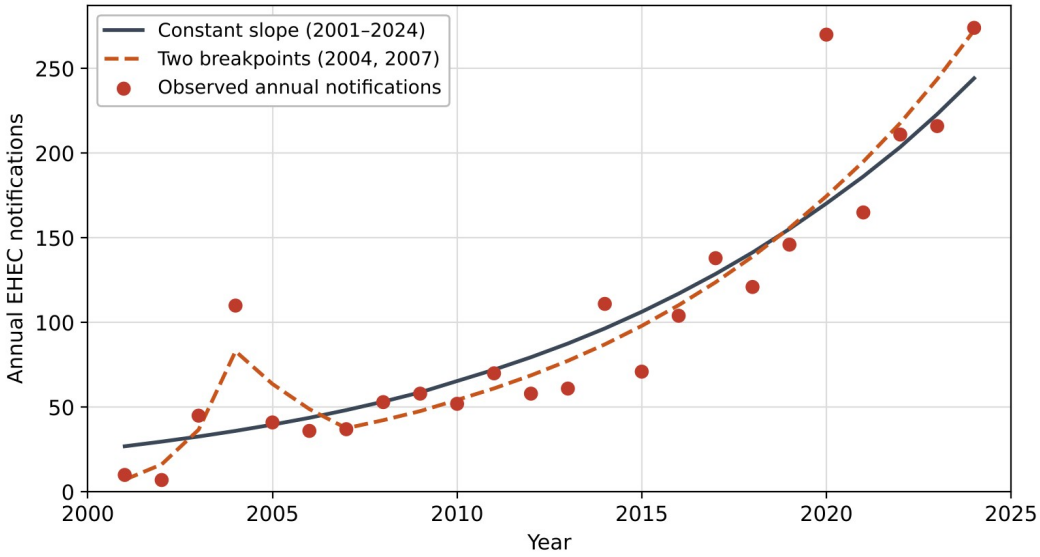
Supplementary Figure S1. Breakpoint detection power



Power to detect a single negative-binomial breakpoint in the 24-year annual series at the dispersion used for the simulation (α = 0.16; the corrected empirical value, 0.155, is immaterial at this precision); 1,000 simulations per effect size; detection criterion ΔBIC < −6. (A) Continuous post-breakpoint additional slope change; detection is approximately 94% for ≥25%/year (100% at ≥50%/year). (B) Single-year multiplicative step shock; detection remains low (about 40%) even for a five-fold spike. Dashed line = 80% power; dotted line = 5% nominal false-positive rate.

Supplementary Figure S2. Constant-slope and two-breakpoint negative-binomial fits

Supplementary Figure S2. Constant-slope and two-breakpoint negative-binomial fits



Observed annual EHEC notifications (points) with the constant-slope negative-binomial GLM fit (solid line) and the two-breakpoint fit with breaks at 2004 and 2007 (dashed line), both with log-population offset at the empirical dispersion α = 0.155. The two-breakpoint model met the pre-specified criterion (ΔBIC = −11.0; Supplementary Table S9); it describes a transient excess around 2004 followed by a log-linear rise from 2007 (segment slope +12.1% per year), and no breakpoint was supported within 2007–2024.